

Project Title: Meal Planning and Recipe Management Web Application

Module: COM661 – Full-stack Strategies and Development

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1. Introduction

The rapid growth of health awareness and digital solutions has increased the demand for applications that assist users in managing their dietary habits efficiently. This project presents the design and development of a **Meal Planning and Recipe Management Web Application**, aimed at providing users with a comprehensive platform to manage recipes, plan meals, and track nutritional intake.

The application is built using a modern full-stack architecture, integrating an Angular front-end, a Flask-based backend, and a MongoDB database. It supports dynamic user interaction, real-time data updates, and role-based access control for administrators and standard users.

2. Objectives

Aim

To design and develop a scalable, interactive, and user-friendly web application for managing recipes and meal planning with integrated nutritional tracking.

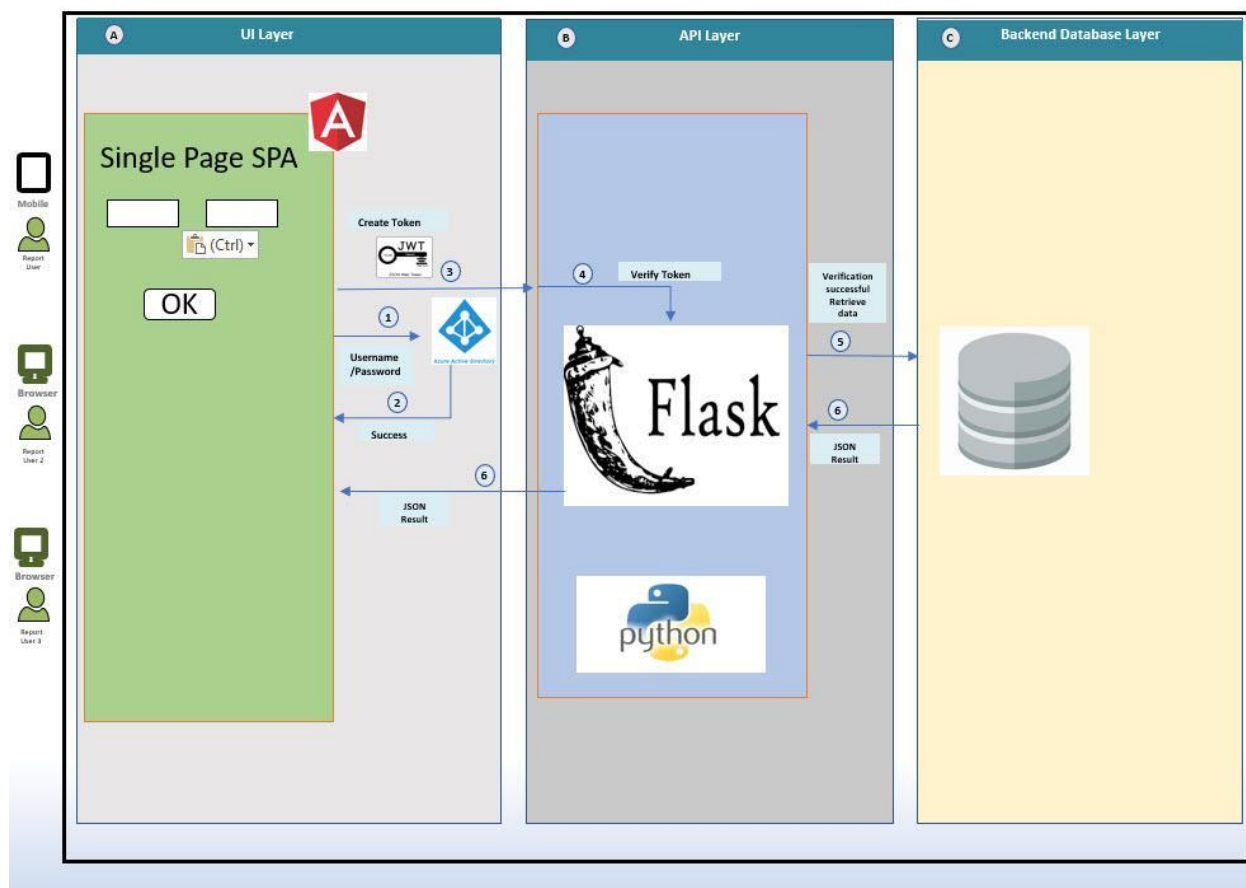
Objectives

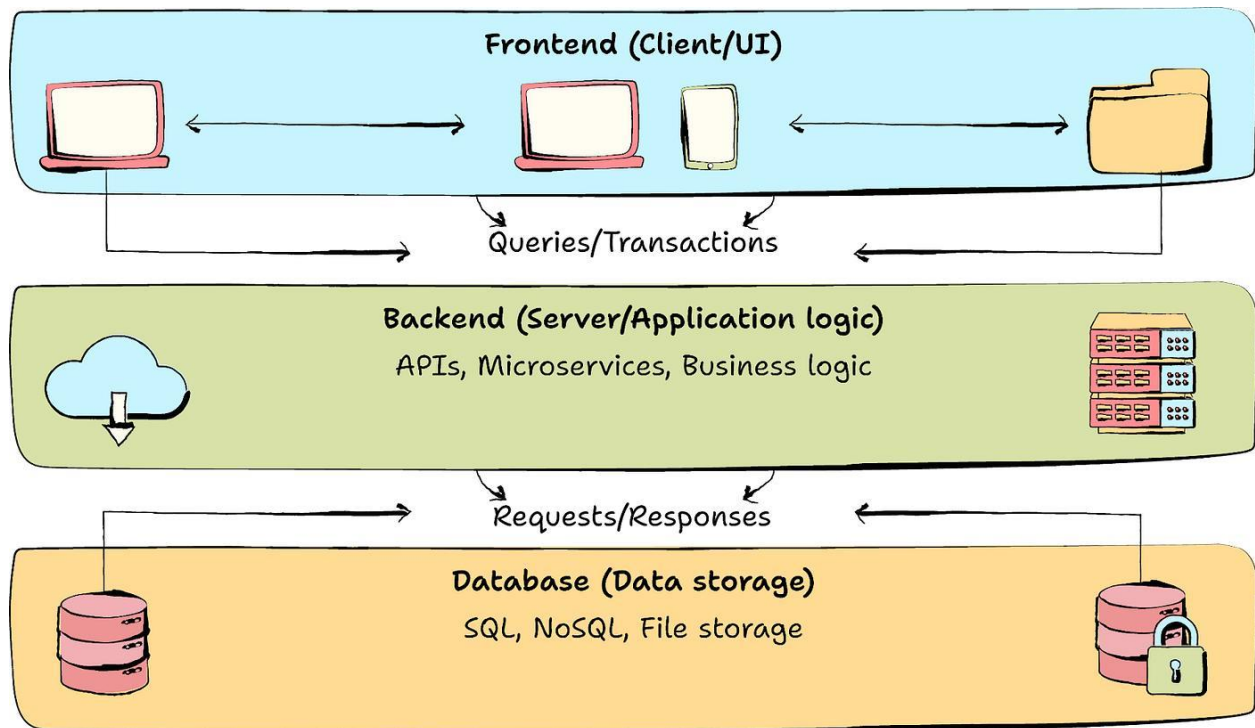
- Develop a responsive front-end using Angular
- Implement RESTful APIs using Flask
- Store and manage data using MongoDB

- Enable CRUD operations for recipes and meal plans
- Implement authentication and role-based access control
- Provide nutritional analysis and tracking features
- Design a modern and responsive UI

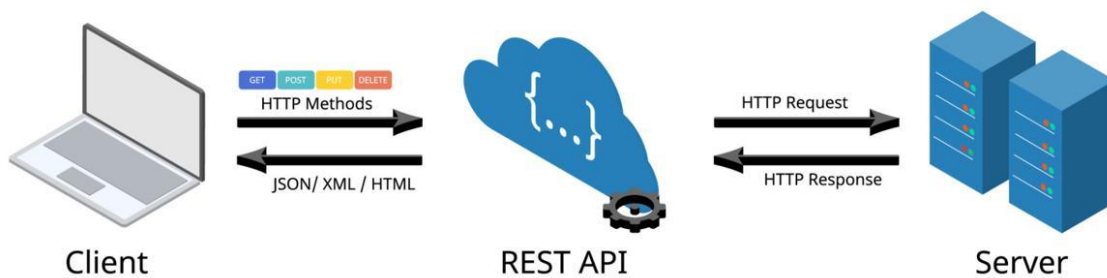
3. System Architecture & Design

3.1 Architecture Overview





REST API Model



Presentation Layer (Frontend)

- Built using Angular
- Handles UI rendering and user interaction
- Uses components and routing

Application Layer (Backend)

- Built with Flask
- Processes HTTP requests
- Contains business logic

Data Layer (Database)

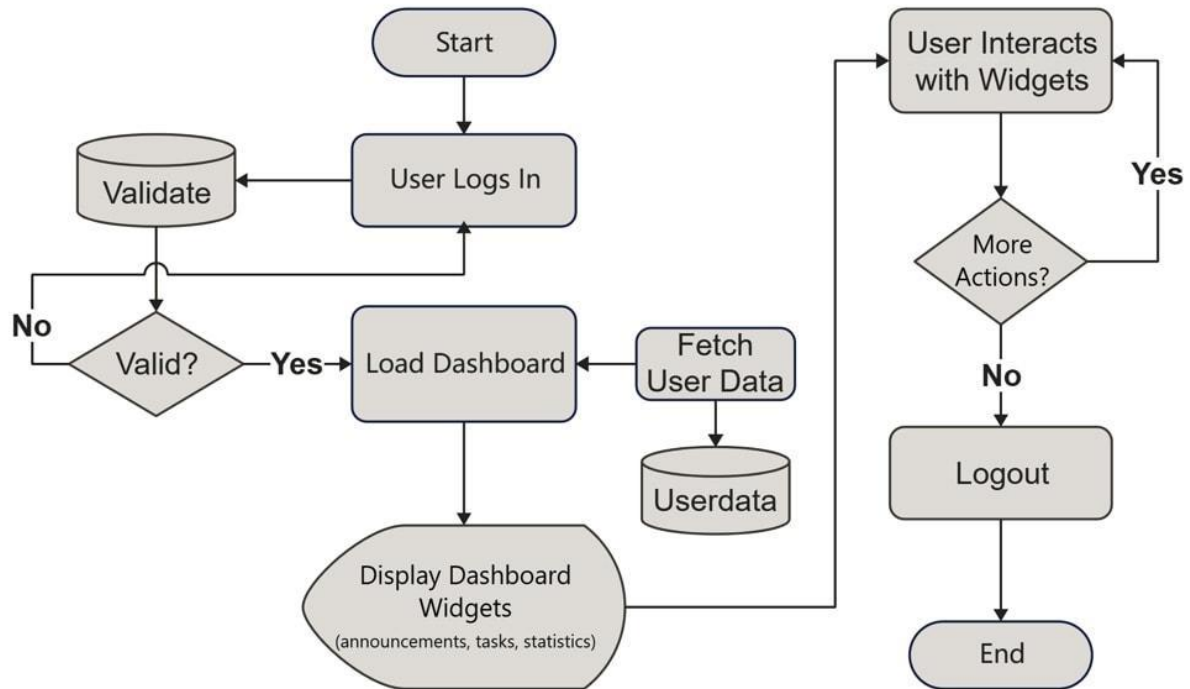
- MongoDB stores data in JSON-like documents
- Handles recipes, meal plans, and user data

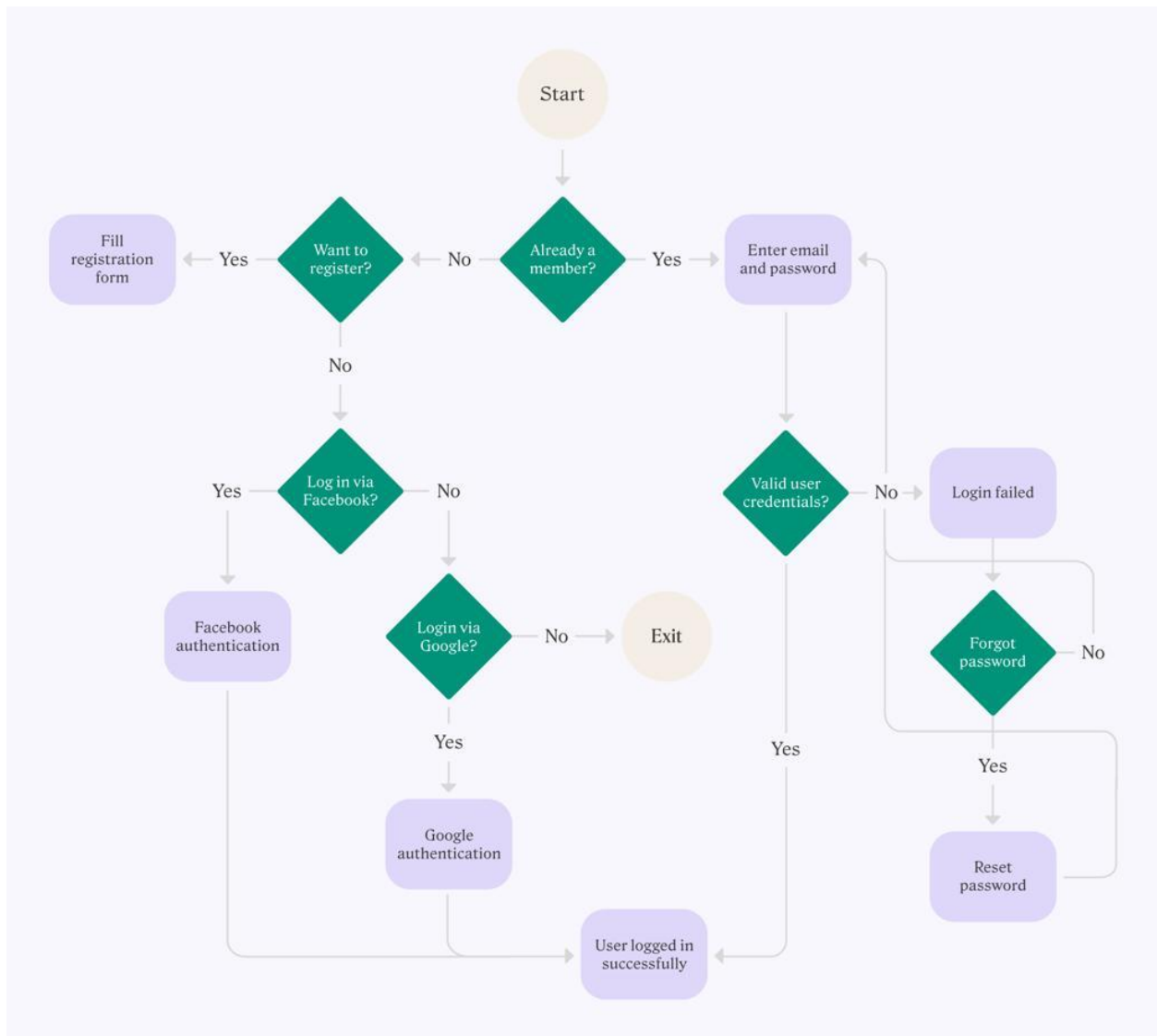
3.2 Data Flow

1. User interacts with the Angular interface
2. Angular sends HTTP request via service
3. Flask API processes request
4. MongoDB performs database operation
5. Response returned to Angular
6. UI updates dynamically

4. Application Flow (User Journey)

User Flowchart





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Flow Description

- User opens application
- Navigates through dashboard
- Selects features (recipes, meal plans, nutrition)
- Data is fetched and displayed dynamically

5. Admin Access and Role-Based Control

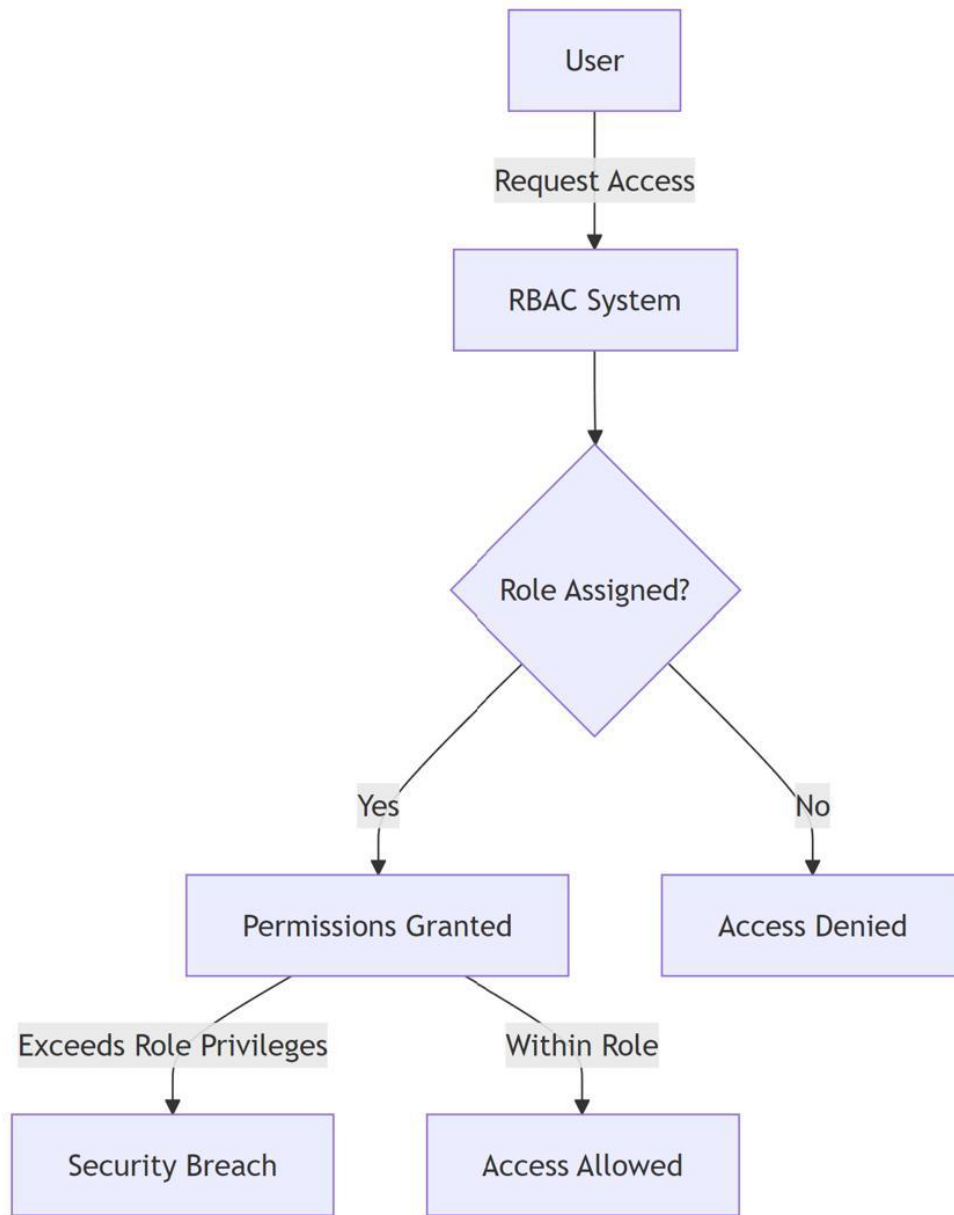
5.1 Admin Credentials (Demo)

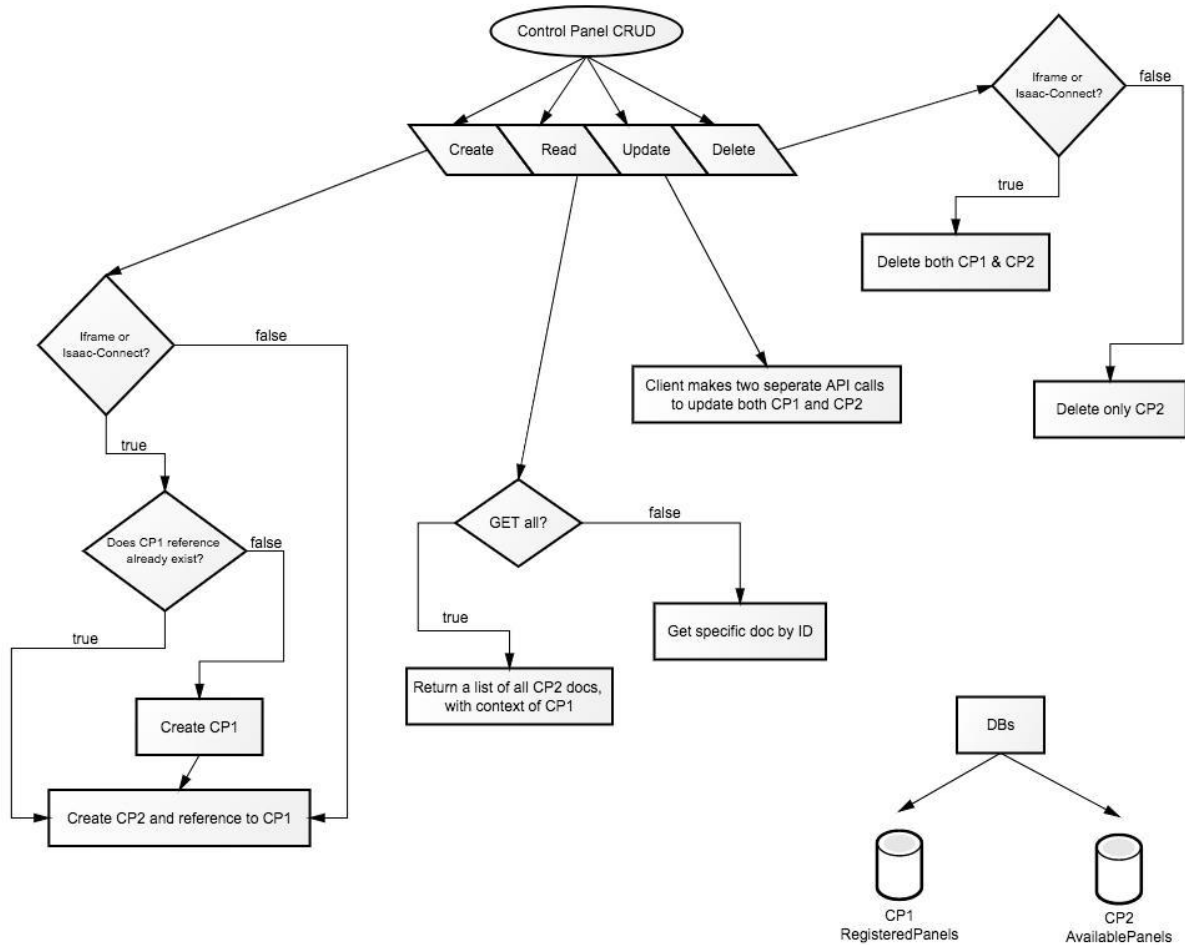
- Username: admin2
- Password: admin@123

5.2 Admin Capabilities

- Create, update, delete recipes
- Manage meal plans
- Access full system data

5.3 Admin Flow





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Implementation

- Role stored during authentication
- Angular conditionally displays admin features
- Backend validates operations

6. Technologies Used

Frontend

- Angular (TypeScript, HTML, CSS)

- Component-based architecture
- Angular routing

Backend

- Flask (Python)
- RESTful APIs

Database

- MongoDB

Authentication

- Auth0
- Custom login system

Additional

- LocalStorage
- CSS animations

7. Features Implementation

7.1 Recipe Management (CRUD)

How it Works

- User submits form → Angular sends POST request
- Flask processes data → stores in MongoDB
- Data retrieved via GET requests
- UI updates dynamically

7.2 Meal Planning

- Users create structured meal plans
- Plans stored in MongoDB
- Retrieved and displayed dynamically

7.3 Recipe Instructions as Arrays

- Instructions stored as arrays
- Displayed using Angular loops (*ngFor)
- Allows flexible formatting

7.4 Image Support

- Images stored as URLs
- Rendered directly in UI
- Reduces server complexity

7.5 Nutrition Tracking

Users input:

- Calories
- Protein
- Carbohydrates
- Servings

Processing

- Values stored with recipes
- Calculations performed dynamically

7.6 Dynamic Recipe Suggestions

- Recipes filtered (e.g., high protein)
- Angular filters dataset in real time
- No page reload required

7.7 Nutrition Tracker (Advanced Feature)

Users input:

- Age
- Weight
- Height
- Gender

Calculation

- BMR formula applied
- Macronutrients calculated

Storage

- Data stored in localStorage

7.8 Dynamic Pie Chart

- Built using CSS conic gradients
- Converts nutrient percentages into visual segments
- Updates dynamically

8. User Interface Design

Hello, admin2

Home

Logout



Plan Your Meals Smarter

Organise recipes, track nutrition, and simplify your weekly meals.



Recipes

Browse your recipe collection and manage saved recipes.

[View Recipes](#)



Mealplans

View your weekly meal plans and planned meals.

[View Mealplans](#)

Activate Windows

Go to Settings to activate Windows.

Hello, admin2

Home

Logout

Nutrition Tracker

Estimate your daily calorie and macronutrient intake targets.

Age

021

Weight (kg)

089

Height (cm)

0180

Gender

Female

Activity Level

High activity

[Calculate Targets](#)

Daily Targets



3323
Calories

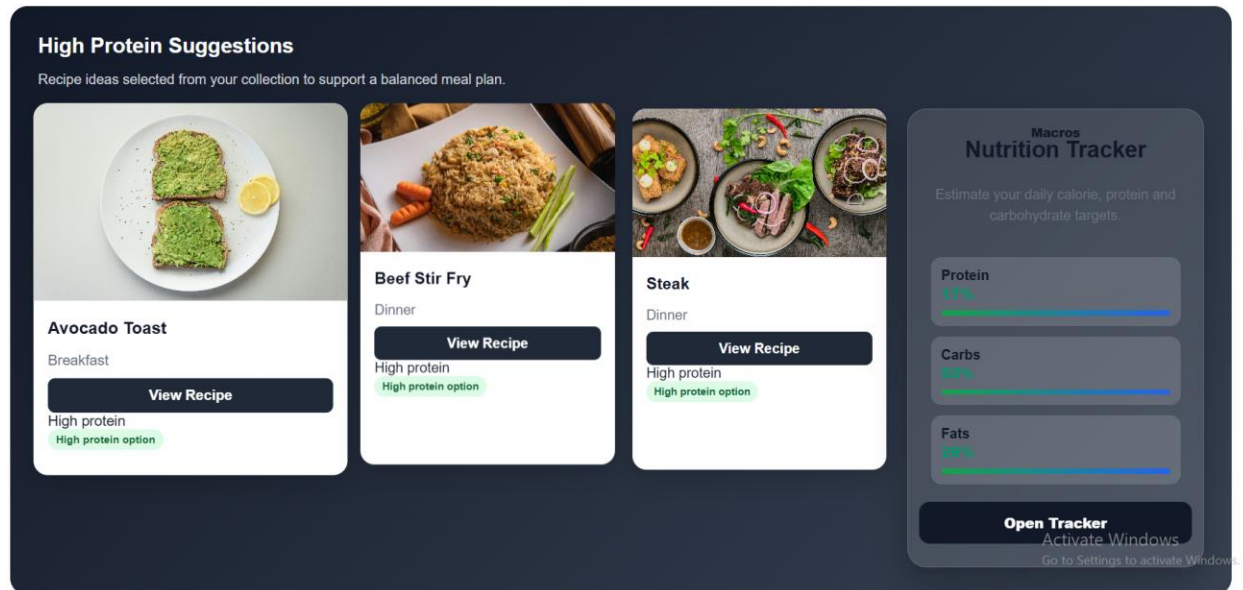
142g
Protein

374g
Carbs

92g
Fats

Activate Windows

Go to Settings to activate Windows.



Design Features

- Dashboard-style homepage
- Hero section with image
- Card-based layout
- Hover animations
- Responsive design

Implementation

- Angular templates and binding
- CSS Grid/Flexbox
- Media queries for responsiveness

9. User Guide

How to Run

1. Extract project ZIP
2. Open terminal

3. Run:
 npm install
4. Start app:
 ng serve
5. Open: <http://localhost:4200>

How to Use

- Navigate dashboard
- Add recipes
- Create meal plans
- Use nutrition tracker

10. Developer Guide

- Angular handles UI and routing
- Services handle API calls
- Flask handles backend logic
- MongoDB stores data
- Code structured for scalability

11. Challenges Faced

- Managing asynchronous API calls
- Implementing authentication
- Designing responsive UI
- Structuring NoSQL data

12. Future Improvements

- JWT authentication
- Cloud deployment
- Advanced analytics
- Real-time updates

13. Conclusion

The project successfully demonstrates a full-stack web application integrating Angular, Flask, and MongoDB. It delivers a feature-rich platform for managing recipes and meal plans while incorporating advanced functionalities such as nutritional tracking and role-based access control. The system reflects strong design principles, scalability, and practical implementation of modern web technologies.